Project Title: MergeMap: A Network Analysis of Model Merging on Hugging Face

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I. Motivation

Hugging Face hosts over three million public AI models. Many are fine-tuned, quantized or adapted from other models, forming family trees. A *merged* model is different. It combines the weights of two or more parents, so merges can join separate families, such as Llama and Mistral. This matters for AI supply-chain risk: a license or safety problem in one model can spread to all models that inherit from it.

II. Related Work and Limitation

Laufer et al. [1] built family trees for about two million Hugging Face models and studied how traits pass from parent to child. Wei et al. [2] identified disruptive models in the open-source LLM community. Horwitz et al. [3] mapped the Hub as a model atlas (a DAG) and found merged LLMs gain more downloads. These works treat lineage as trees, or merges as one model type among many. None study the merge network itself: which families merges connect, how often close relatives merge, and how merges change the reach of a flawed base model. We address this gap with four research questions:

RQ1. How much do merges connect separate model families?
RQ2. Who merges with whom, and how often do close relatives merge?
RQ3. Which models act as bridges or hubs?
RQ4. How many models inherit a flawed base model, and how much do merges add?

III. Data Set

We use the public `cfahlgren1/hub-stats` dataset (`models.parquet`, Apache-2.0), a daily dump of Hugging Face model metadata; we fix one dated snapshot for reproducibility. Each model lists its declared base model(s) and the relation type: *finetune*, *adapter*, *quantized* or *merge*.

IV. Graph Model

Node = model; directed edge parent $\rightarrow$ child, labeled by relation type. After cleaning, the graph is a DAG. Merges with one known parent count for reach only. Each model gets a *family* label from its root base model(s). We study the graph with and without merge edges.

V. Metrics and Algorithms

- **Structure (RQ1):** degree distributions with power-law fits, weakly connected components, and path lengths [4], [5].
- **Mixing (RQ2):** a cocitation network of merge parents, family assortativity, a family-to-family merge matrix, and an inbreeding rate (parents sharing an ancestor within 1–3 generations).
- **Centrality (RQ3):** in-degree, PageRank, approximate betweenness, and cocitation communities vs. family labels.
- **Risk reach (RQ4):** number of descendants of each base model, with and without merges, and the effect of removing top models.
- **Baseline:** degree-preserving random graphs, to test if mixing and inbreeding patterns are real.

We implement reach (BFS), family labeling and inbreeding search in Python; standard metrics use `python-igraph`.

VI. Expected Knowledge

We expect heavy-tailed degrees (RQ1) with a few base models as hubs (RQ3), a much larger component with merges (RQ1), mostly within-family merging with few cross-family bridges (RQ2), and higher risk reach through merges (RQ4). This shows how far one flawed model can spread, and which models to audit first.

VII. Expected Challenges

- **Incomplete and noisy metadata:** many models declare no parent, so counts are lower bounds; we drop self-loops, missing models, and parents newer than children.
- **Scale:** exact betweenness is slow; we sample.
- **Family labels:** merges mix families, so labeling needs a clear rule.
- **Data size:** the raw file has about three million rows; we submit the filtered edge list and a download script.

VIII. Timeline

Nov 11: proposal, graph built. **Nov 12–26:** metrics and algorithms. **Nov 27:** demo of steps 1–4. **Dec 7:** report, code and data.

References

[1] B. Laufer, H. Oderinwale, and J. Kleinberg, "Anatomy of a machine learning ecosystem: 2 million models on Hugging Face," arXiv:2508.06811, 2025.
[2] X. Wei, L. Kang, X. Pan, and J. Yang, "Identifying disruptive models in the open-source LLM community," arXiv:2604.11618, 2026.
[3] E. Horwitz, N. Kurer, J. Kahana, L. Amar, and Y. Hoshen, "Charting and navigating Hugging Face's model atlas," arXiv:2503.10633, 2025.
[4] A.-L. Barabási, *Network Science*. Cambridge, U.K.: Cambridge Univ. Press, 2016.
[5] M. E. J. Newman, *Networks: An Introduction*. Oxford, U.K.: Oxford Univ. Press, 2010.